
Stress Management & Burnout Prevention for Medical Laboratory Professionals

Protecting the Brains Behind the Diagnostic Backbone

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- *Thursday, June 25, 2026*



Objectives

What We Will Cover

- **Define:** Differentiate daily workplace stress from the clinical syndrome of burnout.
- **Review:** Analyze recent peer-reviewed evidence on the lab workforce crisis.
- **Identify:** Highlight the systemic and individual triggers within the clinical laboratory.
- **Action:** Introduce evidence-based solutions for lab staff and laboratory leadership.



The State of the Lab: The Diagnostic Backbone in Crisis

The Invisible 70%: Lab professionals drive up to **70% of clinical decisions** but remain invisible to the public.



The Statistics:

Over **80% of laboratory staff** report active burnout symptoms.

Only **12% of laboratory scientists/technicians** state they are "extremely likely" to stay in the diagnostics field long-term.

Only **35% of lab scientists/techs** feel respected within their broader healthcare organization.

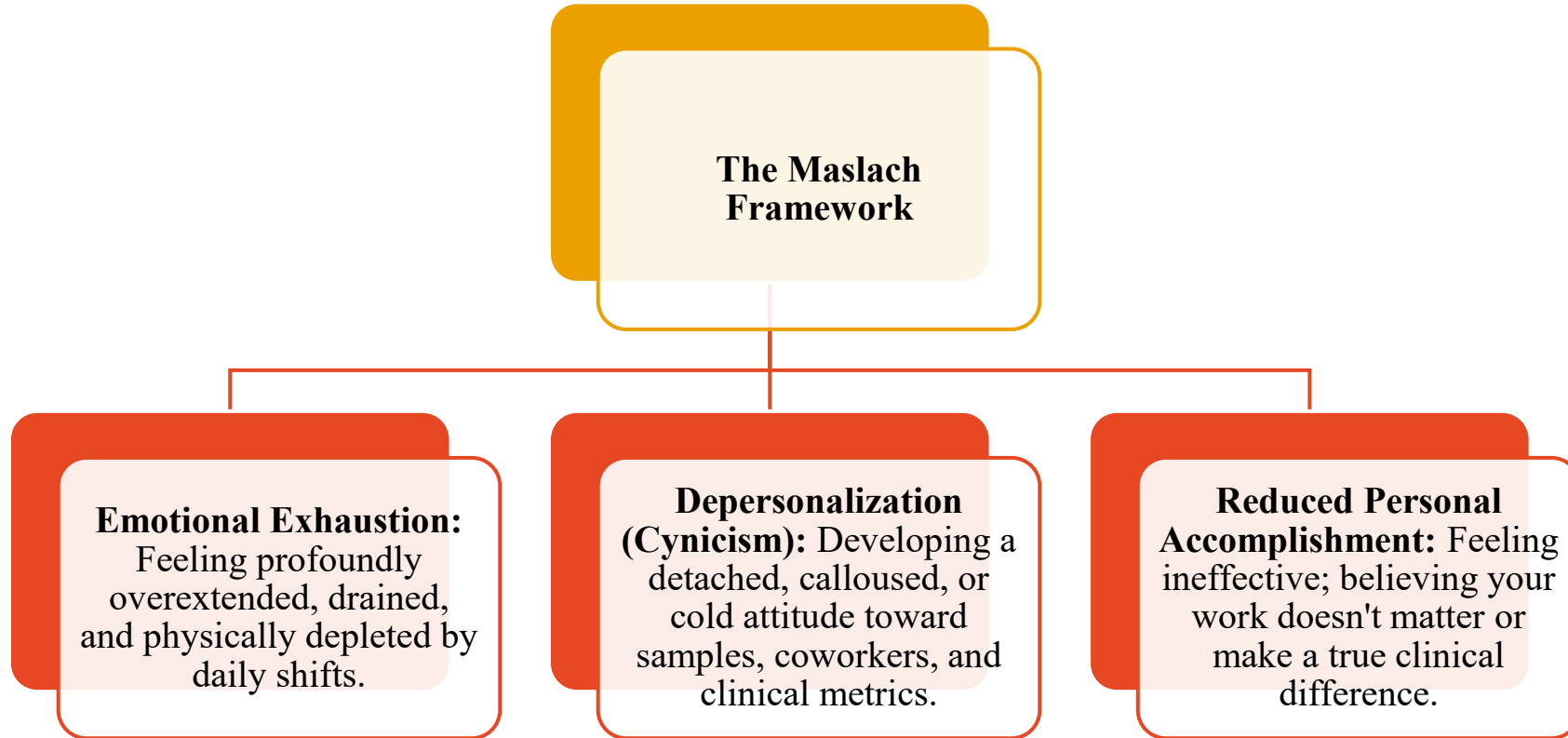
Stress vs. Burnout

- **Stress:** Feeling overwhelmed but believing that if everything gets under control, you will feel better.
- **Burnout:** Feeling empty, devoid of motivation, and beyond caring. It is a state of prolonged disengagement.

Understanding the Threshold

Feature	Workplace Stress	Occupational Burnout
Characterized by	Over-engagement and urgency	Disengagement and blunting
Primary Emotion	Anxiety and physical hyperactivity	Helplessness and hopelessness
WHO Definition	Short-term response to challenges	ICD-11 Syndromic Classification
Impact on Work	Diminished short-term energy	Loss of personal ideals/values

The Three Dimensions of Burnout



Literature Review



The Workforce Shortage Vicious Cycle

- The Supply Gap: The Bureau of Labor Statistics projects a need for 10,000 new laboratory professionals annually, but academic programs produce only 5,000 graduates per year.
- The Aging Workforce: Over 60% of active laboratory workers are rapidly approaching retirement age.
- The Vicious Cycle:

Understaffing----→ Increased Individual Workload-
-----→ Accelerated Burnout--→ High Turnover ---→
Worse Understaffing

Literature Review (Part 2)

The Impact of Stress and Burnout

Impact on Diagnostic Safety & Patient Care

- Error Correlation: High-burnout units display 2.3 times higher odds of adverse clinical events.
- Cognitive Impact: Proactive studies demonstrate that burnout degrades cognitive performance, psychomotor precision, and protocol adherence.
- The Risk: Fatigue can cause errors in sample preparation, barcode tracking, and preliminary slide categorization.
- Perception: 60% of lab staff firmly believe that workplace burnout directly impacts the quality of patient care.



Literature Review (Part 3)

Impacts on Financial and Operational Costs

The operational and financial hit of burnout and turnover is severe:

High Turnover Rates: Medical laboratory technologists experience turnover rates of roughly 15.9%. The cost to replace a single clinical lab technician or lab scientist can reach up to twice their annual salary, including onboarding and productivity losses.

Systemic Financial Loss: Turnover-related expenses cost the average hospital millions annually due to lost billings, overtime premiums, and premium staffing (e.g., traveling techs).

Disengagement Expenses: Beyond turnover, employee disengagement and burnout can cost employers between \$3,999 and \$20,683 per disengaged worker annually.



Unique Triggers in the Clinical Laboratory

Why Laboratory Medicine is Different



The "Invisible" Nature: Rarely interacting with patients leaves staff isolated from the direct human impact of their work.

The "Zero-Error" Expectation: Absolute perfection is expected on every shift; there is no margin for error.

The Shift-Work Grind: 24/7/365 testing requirements lead to disruptive night shifts, erratic schedules, and poor sleep architecture.

The "STAT" Environment: Unrelenting pressure from emergency rooms and critical care units for rapid turnaround times.

Administrative Burden: Strict compliance, QA/QC documentation, regulatory requirements, and Laboratory Information System (LIS) failures.

Frameworks for Prevention

A Dual-Action Approach

- **Organizational Changes:** Systemic shifts always achieve a 2x to 4x greater impact on burnout than individual wellness strategies alone.
- **Individual Interventions:** Personal coping skills function as vital buffers to preserve daily health while systemic issues are resolved.
- **The Goal:** Merging top-down structural modifications with bottom-up individual boundary-setting.



Organizational Interventions

Re-engineering the Lab Environment



Leadership's Role

Address burnout is a strategic priority, not an HR afterthought.

Destigmatize mental health issues in the laboratory.

Organizational Interventions

Workflow Optimization: Use automation and AI to reduce daily friction, repetitive manual tasks. 89% of lab staff agree automation is required to sustain current sample volume.

Fair Scheduling: Ensure proper lunch breaks, limiting weekend shifts, and providing manageable work life boundaries. Optimize night shift rotation blocks to stabilize biological sleep architecture. Influence factors of work stress of medical workers in clinical.



Measuring and Managing

Use the Maslach Burnout Inventory (MBI) to measure and track burnout levels within departments.

Engage frontline voices in workflow redesign and fixing broken processes.

Cultivate Active Appreciation: Implement formal institutional feedback loops to celebrate diagnostic milestones.

Encourage employees to utilize their paid time off.

Foster positive working environment.

Individual Interventions: Safeguarding Your Well-being

- **Rigid Cognitive Boundaries:** Re-frame mandatory breaks as a clinical requirement for patient safety. Define strict work-life boundaries (e.g., not checking emails or taking calls off-shift).
- **The 20-Minute Decompression:** Engage in 20 minutes of deliberate physical movement or exercise post-shift to complete the stress.
- **Task Management:** Communicate clearly with management about workload limitations to establish priorities.
- **Sleep Architecture Protection:** Utilize blackout curtains and noise-canceling tools to protect daytime rest after night shifts.
- **Peer Support:** Form small internal networks to debrief after handling major trauma cases or experiencing equipment failures.
- **The Stress First Aid (SFA):** Apply this model for self-care and peer support framework .
- **Healthy Lifestyle as a Medical Tool**
 - Physical activity: short bursts during the day or longer extended exercise sessions.
 - Nutrition: Fuel the body to sustain cognitive precision and psychomotor skills.
 - Practice Mindfulness: Utilize deep breathing exercises and short micro-breaks during shifts.
 - Seek Support: Take full advantage of Intermountain Health's Employee Assistance Program (EAP).
 - Community: Talk with colleagues about mental health during non-technical staff meetings.



Setting Boundaries

The ability to say no- to set your own boundaries

- Recognize when a boundary is out of alignment (anger, frustration, resentment).
- Define boundary exceptions – patient safety, workflow.
- Identify barriers to upholding a boundary.
- Communicate the boundary to others.
- Keep it simple – you are not required to explain to others why you have a boundary.
- Discuss feelings of success and failure.
- Remind yourself - you have the right to self-care.

Why do we need boundaries?

Why do we say “yes” to something, then regret the decision and blame others?

- We are not compassionate and kind to people who are walking all over us. Why do we allow it?
- We can't be upset with others for not upholding our boundaries
- We must “unlearn” this behavior
- No one can do this for us

What are boundaries?

Boundaries are guidelines, rules or limits that a person creates to identify for themselves what are reasonable, safe and permissible ways for other people to behave around them. Put another way, boundaries are what is okay and not okay for you.

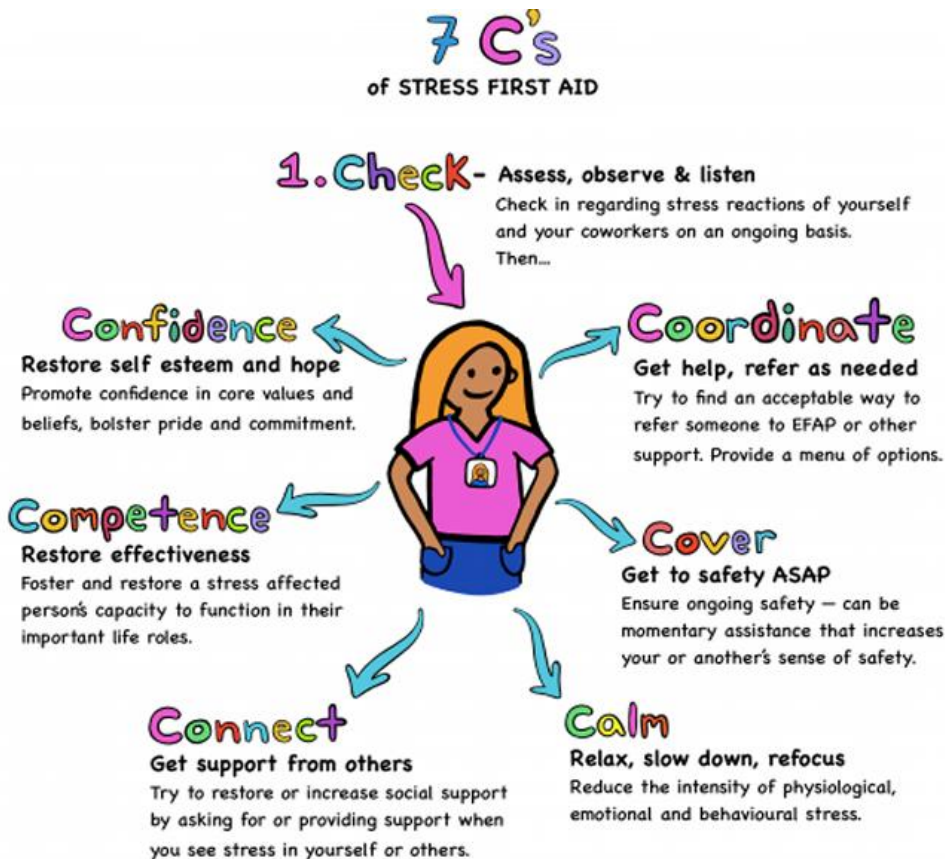
- They are crucial for mental health and wellbeing
- If you feel frustrated, guilty, irritated, or resentment, you may need to set or realign a boundary
- Healthcare, as a culture, teaches us to ignore our personal boundaries (self-care) and we slowly lose the ability to understand why we are miserable
- We are rewarded in this healthcare culture for breaking boundaries and punished for upholding them (seriously!) We feel guilt for not helping others, if we try to hold up a personal boundary

- Yet, we will not be compassionate and kind to people who walk all over the boundaries we have set for ourselves. But surprisingly, it's us that needs to hold the boundary, not anyone else
- Set your boundaries or someone (your job, your work) will set them for you.

What do boundaries look like?

- 1 Scheduling time to perform live well 'take 10 workouts'
- 2 Schedule uninterrupted time to get work done Expect them to decline meetings that interfere with this time
- 3 When on call, then off at another time (required/expected)
- 4 Permission to say “no” to others when they are sick/on PTO/handling other personal needs
- 5 Permission (and the expectation) that multitasking is prohibited (multitasking is NOT a job requirement!)
- 6 Schedule more time as a team; for interaction and sharing of information

Stress First Aid Model and Well-being Resources



Stress Continuum

THRIVING

Definition

- Optimal functioning
- Adaptive growth
- Well-being
- Motivated



ACTIONS

- Practice self-care
- Teamwork
- Appropriate rest

SURVIVING

Definition

- Mild and temporary distress
- Feeling irritable, anxious, or down
- Muscle tension
- Less focused



ACTIONS

- Use Stress First Aid

STRUGGLING

Definition

- More intense and ongoing distress
- Loss of control
- Don't feel like normal self
- Negative thinking
- Feeling overwhelmed



ACTIONS

- Use Stress First Aid
- Seek additional support

IN CRISIS

Definition

- Severe ongoing distress
- Difficulty functioning well
- Hopelessness
- May include mental health conditions, burnout, insomnia, substance abuse



ACTIONS

- Seeking additional support
- Consider mental health resources
- Consider adjustments to work situation

◀ "I've got this"

"Something isn't right"

"I can't keep this up"

"I can't survive this" ▶

Watson P & Westphal R (2020). Stress First Aid for Healthcare Workers, National Center for PTSD | Colorado Healthcare Ethics Resource <https://cohcwcovidsupport.org/>

THRIVING

To maintain:

- Social connection
- Movement/stretching
- Mindfulness
- Humor/play
- Sleep (7-8 hours)
- Proper nutrition
- Hydration
- Boundary-setting
- CredibleMind
- Sworkit App
- Be Well program
- Huddle Up Lift Up flipbook (prompts to support well-being)
- Going home checklist

SURVIVING

Resources for help:

- Everything in **Thriving** section
- Unit leaders
- Employee Assistance Program
1-800-832-7733
eap@imail.org
- Peer Support
- Coach or mentor support
- Mindfulness practices:
*Three Breaths
*5-4-3-2-1 Grounding:
5 things you see
4 things you hear
3 things you feel

STRUGGLING

Resources for help:

- Everything in **Surviving** Section
- Employee Assistance Program
1-800-832-7733
eap@imail.org
- Onsite chaplain/spiritual care
- AskHR
1-801-442-7547

IN CRISIS

Resources for help:

- **IMMEDIATE INTERVENTION**
- Employee Assistance Program
1-800-832-7733
- ED referral/**911**
- Onsite chaplain/spiritual care
- National Suicide Prevention Lifeline **988**
- Behavioral Health Services Navigation Line
1-833-442-2211
- Trevor Lifeline (LGBTQ+):
1-866-488-7386

Peer Support: Caring for ourself and Others

Identify: Pay attention to your team. Be aware of any changes in your coworkers.

Connect: By using phrases like, “I’ve noticed something. Would you like to talk?” or “How are you (really) doing?”

Listen and be present:

Minimize distractions and accept moments of silence.

Empathize: Consider how it might feel to be in that person's situation.

Reflect: On what you are hearing by saying, “Thank you for sharing; it sounds like...”

Respond: By committing to follow up. “I will check in with you... (tomorrow, in a week)?”

Tips and Reminders

- Conversations should be private.
- Be mindful of your own body language and tone of voice.
- Try to put yourself in the other person's shoes to better understand what they are feeling.
- Resist the urge to solve the problem.
- If you are unsure how to respond, it is okay to pause and say, “Let’s connect with AskHR [or EAP or a manager] to get the best resources.”
- Consider any disabilities, language barriers, ethnic, racial or cultural sensitivities.
- Identify and connect to additional support if needed.

Additional Resources

Peer Support: PeerSupport@imail.org

Employee Assistance Program: **1-800-832-7733**

AskHR: **1-801-442-7547**

Behavioral Health Services—Navigation: **1-833-442-2211**

Caregiver Relief: CaregiverAssistance@imail.org

If you or someone you know is experiencing thoughts of suicide or may be in a mental health crisis, call:

Mental Health Crisis Lifeline: **9-8-8**

National Suicide Prevention Lifeline: **1-800-273-8255**

TrevorLifeline (LGBTQ+): **1-866-488-7386**

Summary & Key Takeaways

- Burnout is a systemic issue that arises from problematic workplace environments rather than individual shortcomings.
- There is a significant need for data-driven approaches, as the high rates of burnout in laboratories pose risks to workforce retention and the quality of patient diagnostics.
- Solutions must involve integrating AI and automation with well-defined personal boundaries to effectively address this issue.

In conclusion, burnout should be recognized as a work-related phenomenon rather than a flaw in personal character. Addressing it is crucial for maintaining diagnostic integrity and requires both technological support from management and ongoing self-care from all individuals involved.



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Q&A

Open floor for discussion, sharing coping tips,
and brainstorming laboratory improvements

